

Automated Regression Testing On A Web Application (Test Results Summary)

Demo Assignment: Develop automated test scripts using a test automation framework of your choice. Perform automated regression testing on a web application and analyze the test results. Present your automation approach and discuss the benefits of test automation.

Submit 4 things:

- 1.** Laptop screen recording video - 3-4 min with AI voice notes explaining steps
- 2.** GitHub link - test scripts + README
- 3.** Screenshots - SS of every major step attached, labeled Step 1, Step 2
- 4.** 1-page PDF - test result summary

Create simple automated test scripts, for example a login test where you open the browser, enter details, and check the result. The main goal is to show your approach to automation, run regression tests, and present your work clearly through the video, GitHub link, screenshots, and a short summary report.

Automated Regression Testing On A Web Application (Test Results Summary)

Project: Automated Regression Testing

Website: <https://opensource-demo.orangehrmlive.com>

GitHub Repository:

Tool Used: Selenium WebDriver with Python in VS Code

Objective:

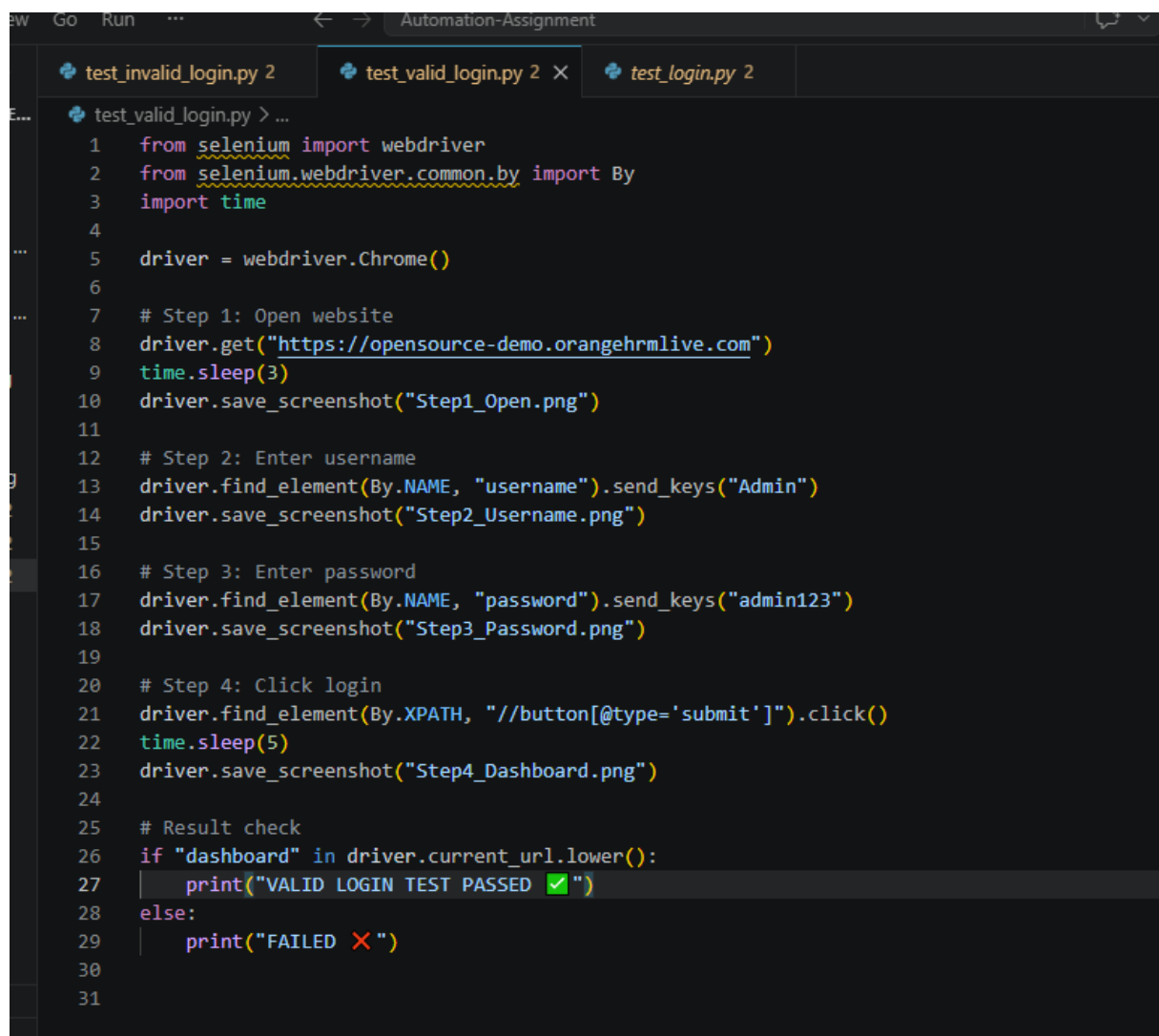
To automate regression testing of login functionality.

Test Scenario: Valid Login Test or Invalid Login Test

TEST CASE 1: VALID LOGIN

CODE:

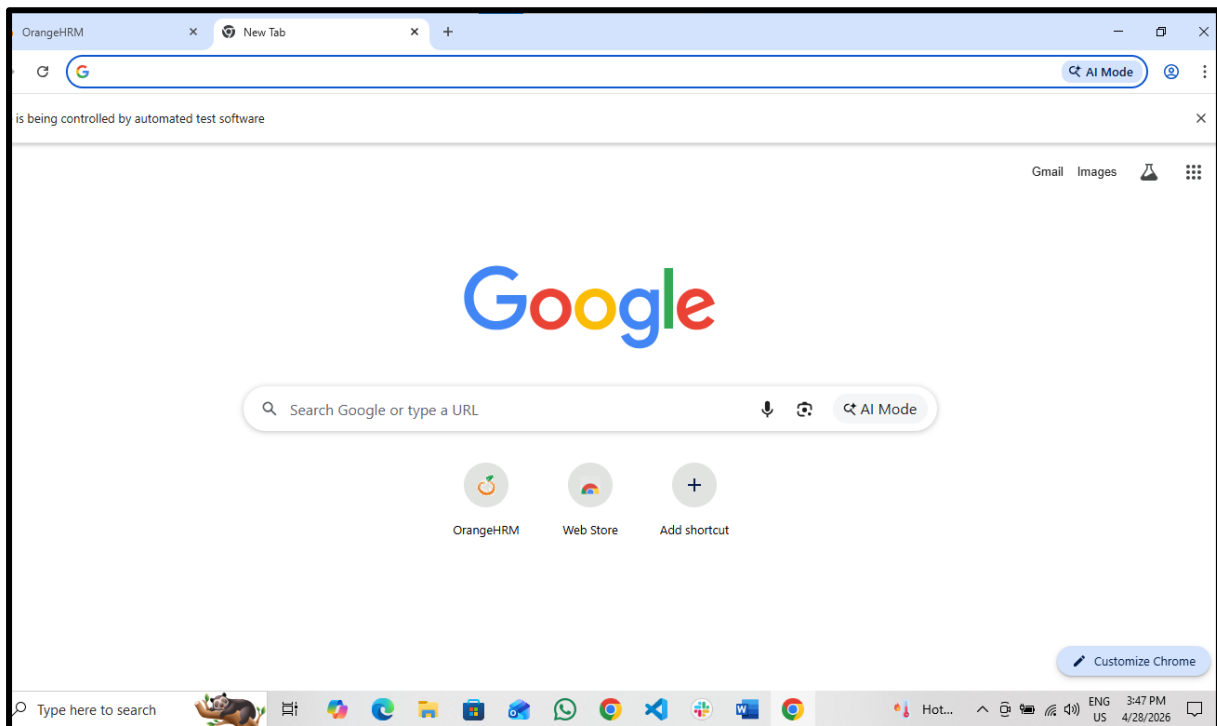
Test_valid_login.py (Automatic Testing):



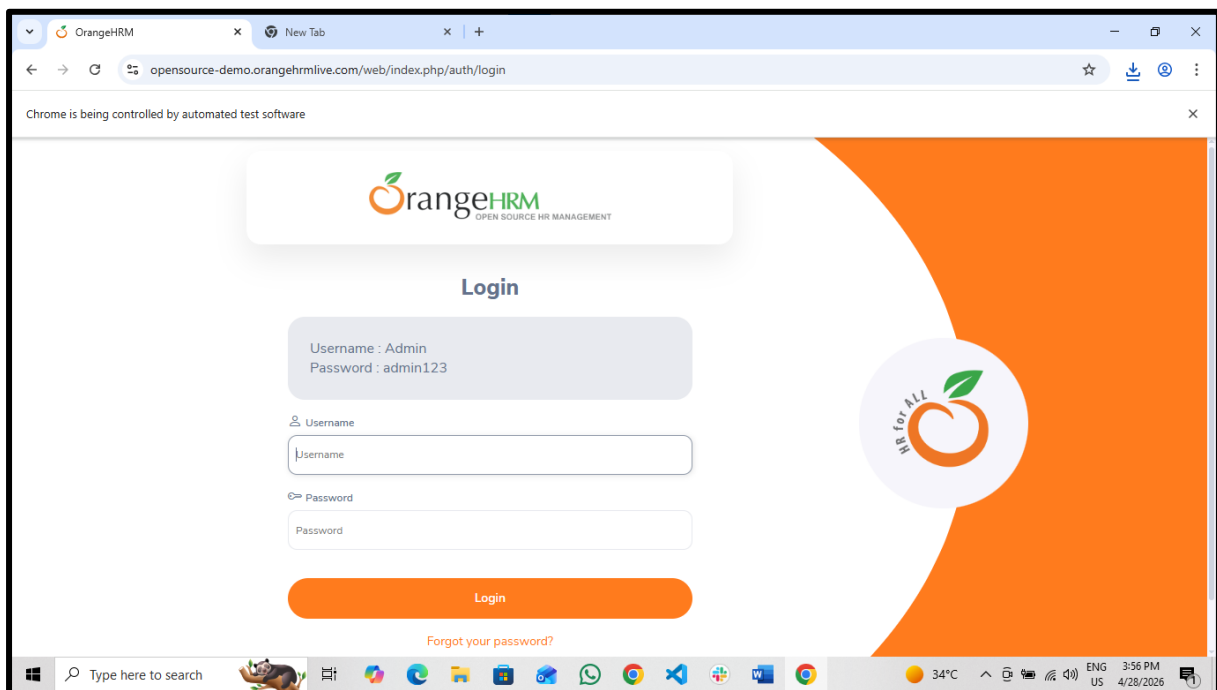
```
Automation-Assignment
test_invalid_login.py 2 test_valid_login.py 2 X test_login.py 2
test_valid_login.py > ...
1 from selenium import webdriver
2 from selenium.webdriver.common.by import By
3 import time
4
5 driver = webdriver.Chrome()
6
7 # Step 1: Open website
8 driver.get("https://opensource-demo.orangehrmlive.com")
9 time.sleep(3)
10 driver.save_screenshot("Step1_Open.png")
11
12 # Step 2: Enter username
13 driver.find_element(By.NAME, "username").send_keys("Admin")
14 driver.save_screenshot("Step2_Username.png")
15
16 # Step 3: Enter password
17 driver.find_element(By.NAME, "password").send_keys("admin123")
18 driver.save_screenshot("Step3_Password.png")
19
20 # Step 4: Click login
21 driver.find_element(By.XPATH, "//button[@type='submit']").click()
22 time.sleep(5)
23 driver.save_screenshot("Step4_Dashboard.png")
24
25 # Result check
26 if "dashboard" in driver.current_url.lower():
27     print("VALID LOGIN TEST PASSED ✅")
28 else:
29     print("FAILED ❌")
30
31
```

OUTPUT

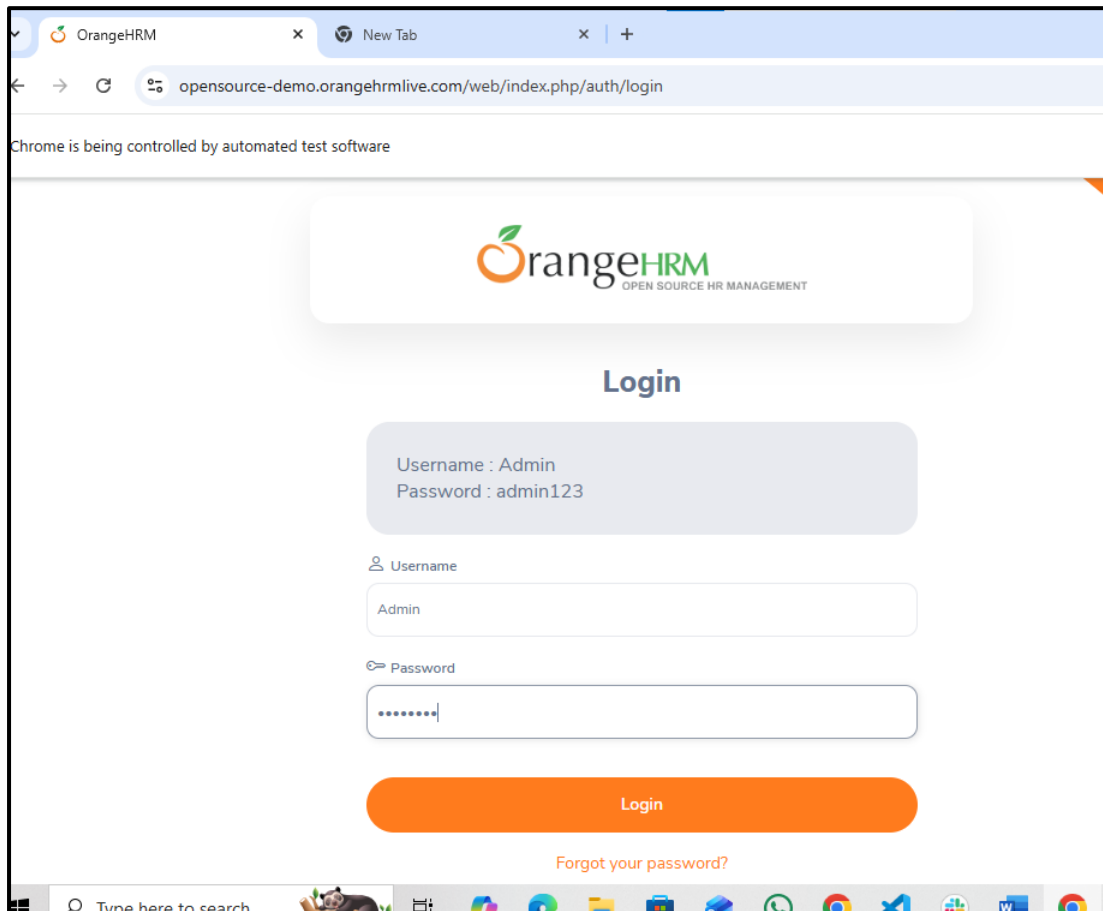
1. Open Chrome browser



2. Open OrangeHRM website



3. Enter Username: Admin and Password: admin123



OrangeHRM

opensource-demo.orangehrmlive.com/web/index.php/auth/login

Chrome is being controlled by automated test software

OrangeHRM
OPEN SOURCE HR MANAGEMENT

Login

Username : Admin
Password : admin123

Username

Admin

Password

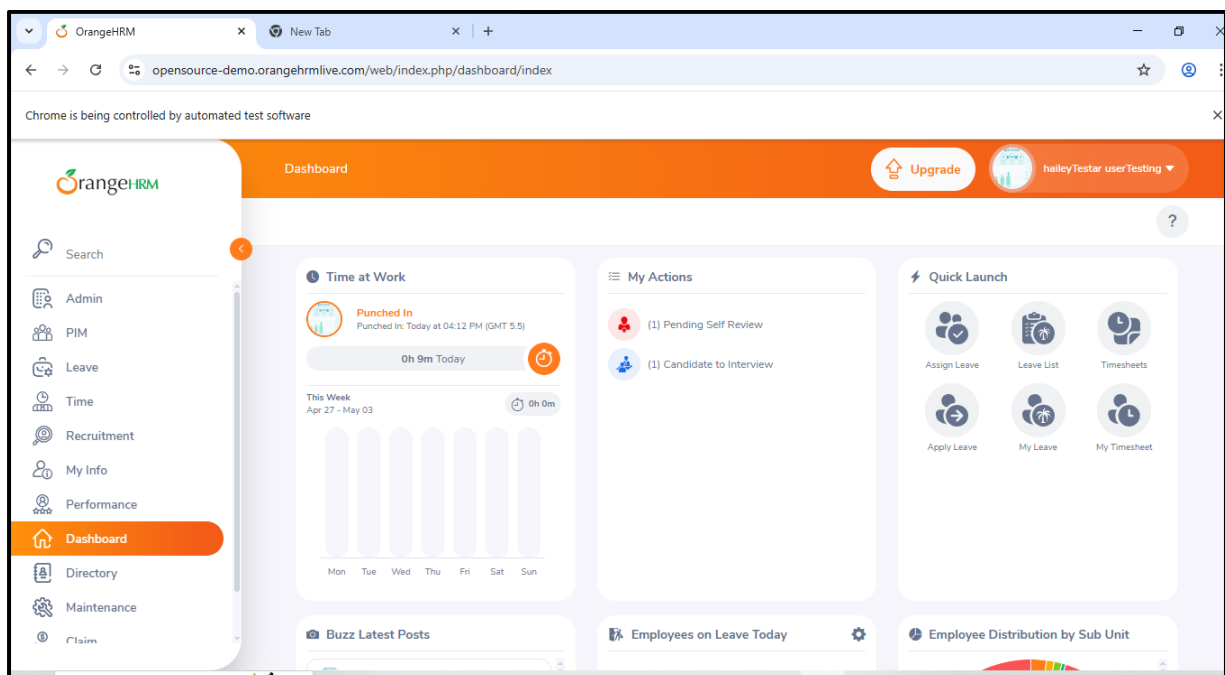
.....

Login

[Forgot your password?](#)

4. Click Login button

LOGIN Successfully



OrangeHRM

opensource-demo.orangehrmlive.com/web/index.php/dashboard/index

Chrome is being controlled by automated test software

Dashboard Upgrade haileyTestar userTesting

Search

- Admin
- PIM
- Leave
- Time
- Recruitment
- My Info
- Performance
- Dashboard**
- Directory
- Maintenance
- Islam

Time at Work

Punched In
Punched In: Today at 04:12 PM (GMT 5.5)

0h 9m Today

This Week
Apr 27 - May 03

Mon Tue Wed Thu Fri Sat Sun

My Actions

- (1) Pending Self Review
- (1) Candidate to Interview

Quick Launch

- Assign Leave
- Leave List
- Timesheets
- Apply Leave
- My Leave
- My Timesheet

Buzz Latest Posts

Employees on Leave Today

Employee Distribution by Sub Unit

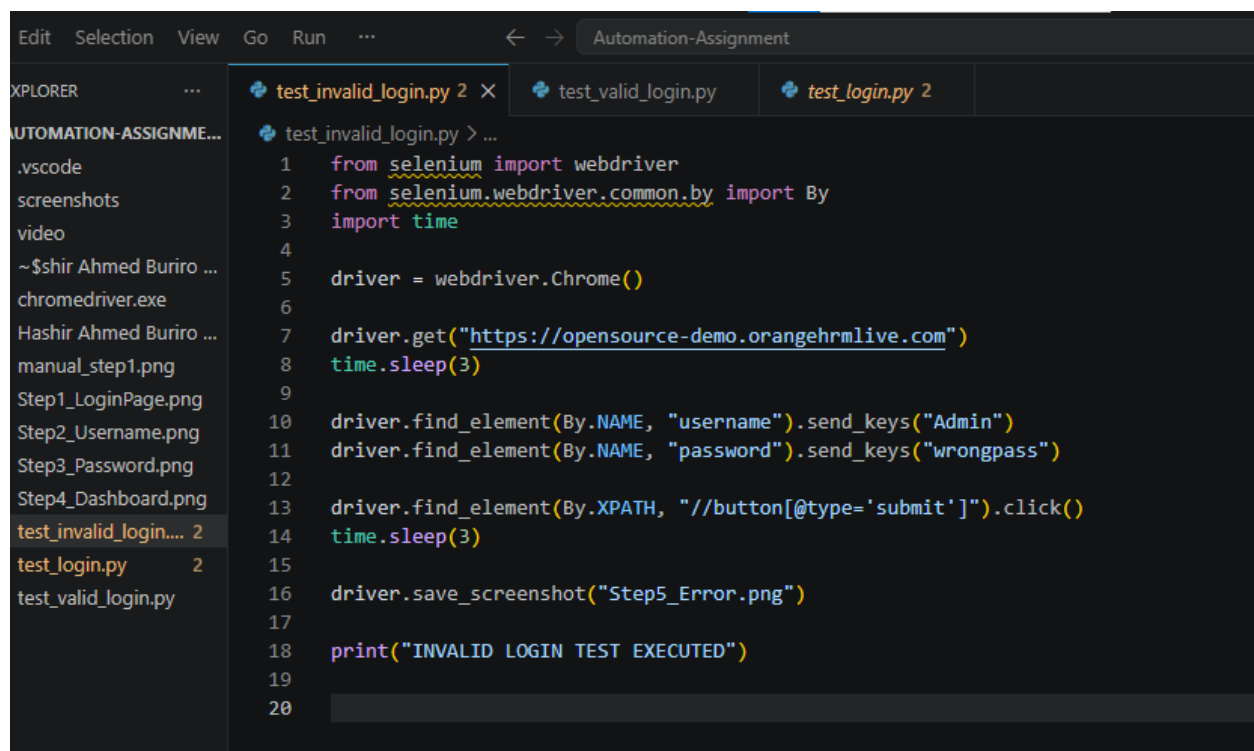
Expected Result:

Dashboard should open successfully

Actual Result:

Dashboard opened successfully

Status: PASS

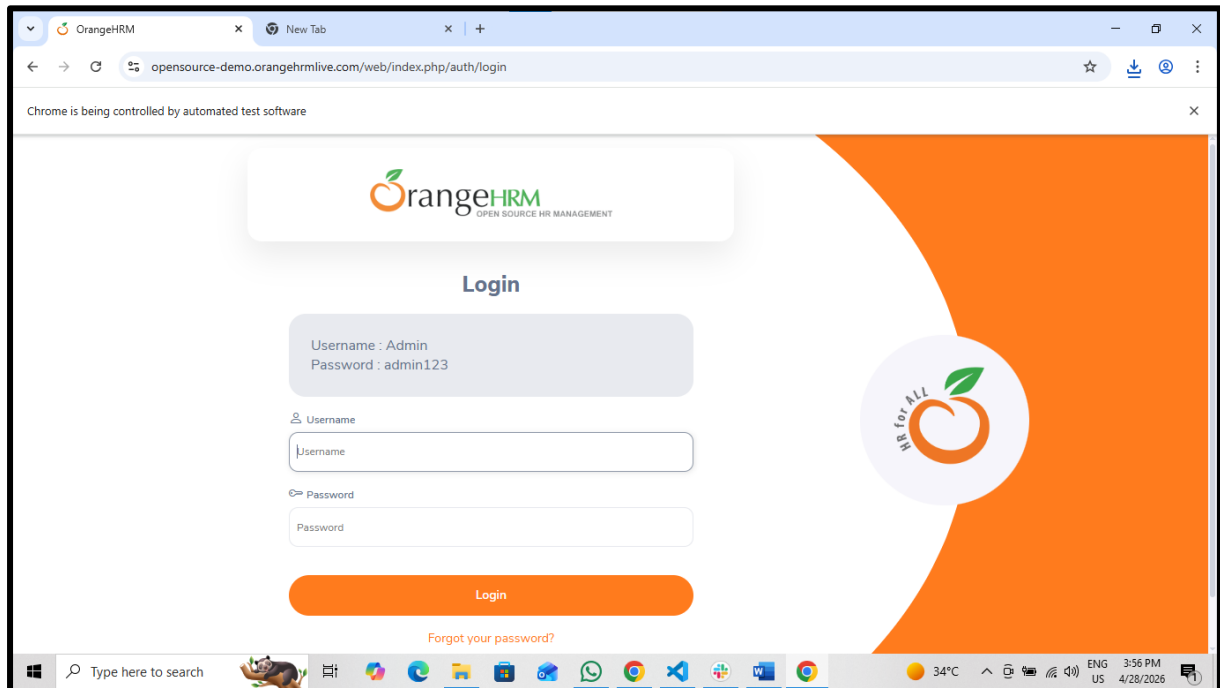
TEST CASE 2: INVALID LOGIN**CODE:****Test_invalid_login.py:**

```
Automation-Assignment
test_invalid_login.py 2 X
test_valid_login.py
test_login.py 2

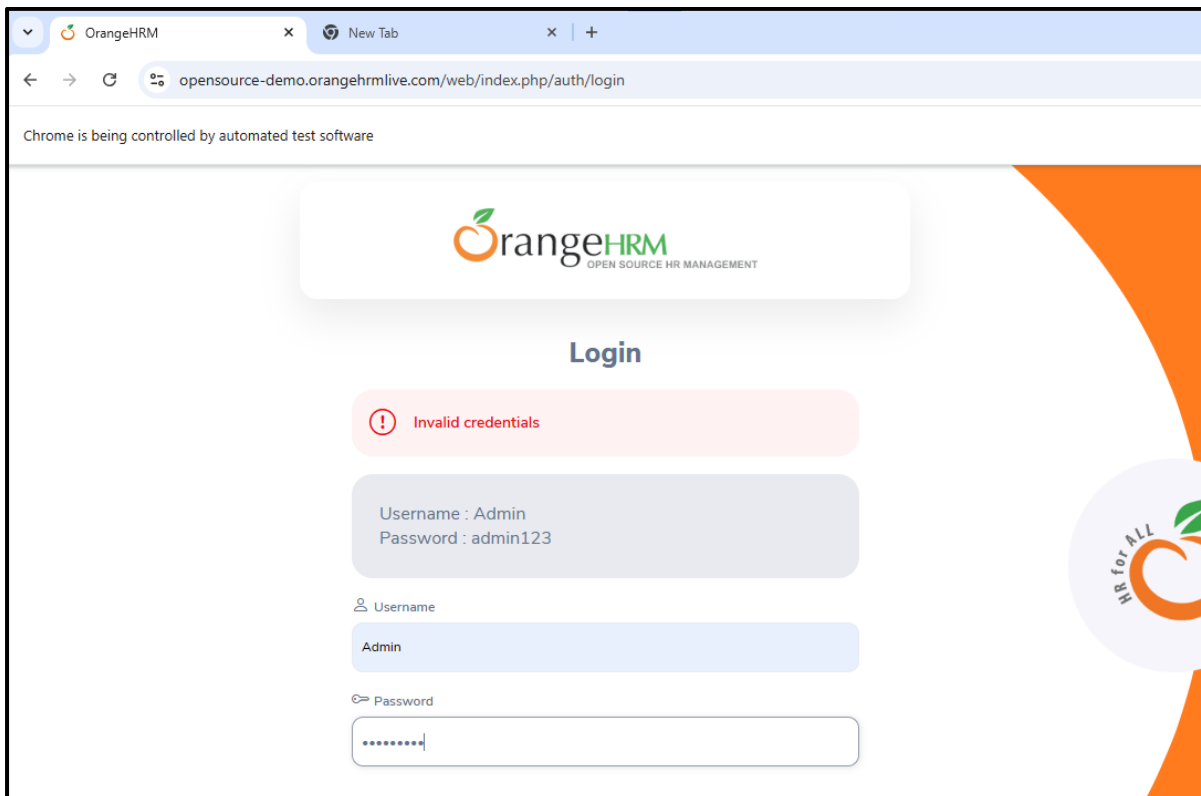
test_invalid_login.py > ...
1  from selenium import webdriver
2  from selenium.webdriver.common.by import By
3  import time
4
5  driver = webdriver.Chrome()
6
7  driver.get("https://opensource-demo.orangehrmlive.com")
8  time.sleep(3)
9
10 driver.find_element(By.NAME, "username").send_keys("Admin")
11 driver.find_element(By.NAME, "password").send_keys("wrongpass")
12
13 driver.find_element(By.XPATH, "//button[@type='submit']").click()
14 time.sleep(3)
15
16 driver.save_screenshot("Step5_Error.png")
17
18 print("INVALID LOGIN TEST EXECUTED")
19
20
```

Output:

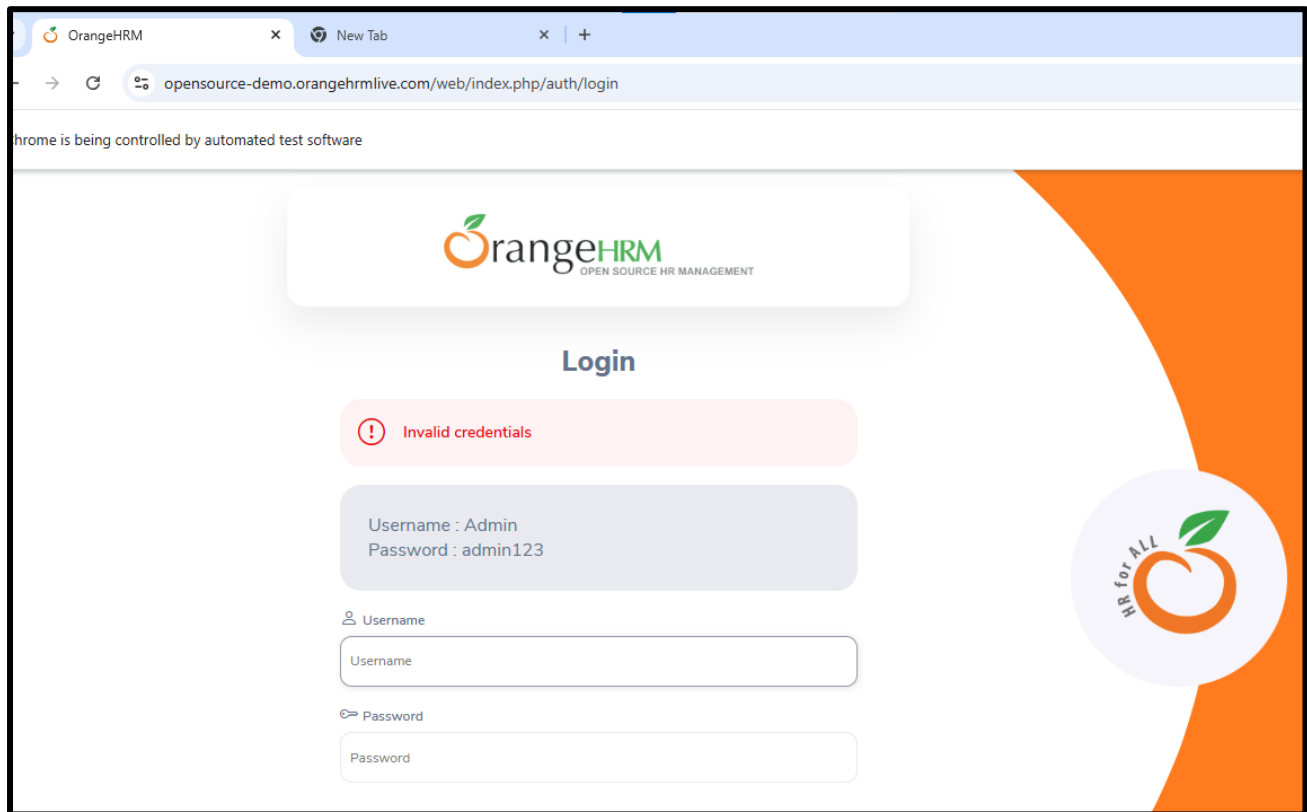
1. Open OrangeHRM website



2. Enter Username: Admin and Enter incorrect password



3. Click Login button

**Expected Result:**

Invalid Credentials

Actual Result:

Error message displayed successfully

Status:

PASS

RESULT:

The automation script executed successfully. The application opened correctly, login credentials were entered automatically, and the dashboard loaded successfully.

Benefits of Manual Testing:

- Verifies real user behavior and user experience
- Helps identify usability and interface issues
- Ensures application functionality works correctly
- Useful for validating both positive and negative test scenarios
- Improves software quality before deployment

CONCLUSION:

Automated regression testing was successfully performed on the OrangeHRM demo web application using Selenium WebDriver with Python. The automation script executed all test steps correctly, including opening the browser, navigating to the website, entering login credentials, and verifying successful login functionality.

The test results confirmed that the application responded correctly to valid login inputs and the dashboard loaded successfully after authentication. The automation process reduced manual effort and ensured accurate and repeatable testing.

This project demonstrated how automation testing improves efficiency, saves testing time, and supports reliable regression testing for web applications.